

Do players correctly estimate what others do? Evidence of conservatism in beliefs

Steffen Huck^{a,*}, Georg Weizsäcker^b

^a Department of Economics, Royal Holloway, University of London, Egham, Surrey TW20 0EX, UK

^b Department of Economics, Harvard University, and Harvard Business School,
Sherman Hall, Boston, MA 02163, USA

Received 9 August 1999; received in revised form 10 November 2000; accepted 10 December 2000

Abstract

In a simple experimental environment a group of subjects was asked to give estimates of a second group's choice frequencies in a set of lottery-choice tasks. The results show that subjects in the first group are on average able to correctly predict the option that is chosen with higher frequency by the second group, but the predictions are systematically inaccurate in that they are distorted toward the uniform prior. Two mechanisms to elicit the expectations were used in the experiment, a quadratic scoring rule and a bidding mechanism. Aggregate results being similar under both mechanisms, the use of the former mechanism consistently yields more accurate predictions. © 2002 Elsevier Science B.V. All rights reserved.

JEL classification: C23; C91

Keywords: Beliefs; Elicitation; Prediction accuracy; Experiments

1. Introduction

In a number of recent studies, experimental economists have attempted to elicit subjects' expectations about the behavior of their opponents in laboratory games. Mostly, these studies are motivated by questions of fairness and reciprocal behavior in contexts of public-goods games and related situations, and the focus of attention lies on correlations between elicited beliefs and the subjects' behavior in the games.¹ Little work, however, has been devoted

* Corresponding author. Tel.: +44-1784-443383; fax: +44-1784-439534.

E-mail addresses: s.huck@rhbnc.ac.uk (S. Huck), weizsack@fas.harvard.edu (G. Weizsäcker).

¹ For example, see the public-goods experiments by Offerman et al. (1996), and by Croson (2000). Beliefs in the trust and related games were elicited by Dufwenberg and Gneezy (2000) and others. A different question motivates the study by Nyarko and Schotter (1999), who use elicited beliefs to describe the learning behavior of subjects in a repeated normal-form game.

to the question whether the subjects' elicited expectations about other subjects' behavior are on average correct, i.e. whether subjects are able to predict others' actions without any systematic bias. Relatedly, the question arises what elicitation mechanisms yield more accurate responses than others.

This paper presents an experimental study in which we asked subjects in one group (Group A) to estimate choice frequencies of a second subject group (Group B) in a given set of binary-lottery-choice tasks. The subjects in Group B did not know about the existence of Group A, and this was known to the subjects in Group A. Thereby, the experimental design excludes issues of coordination or fairness that can arise in other experimental games.² The beliefs of the subjects in Group A about Group B behavior were elicited using two different mechanisms: one in which subjects bid for the right to participate in artificially constructed lotteries (or promissory notes), using a Becker–DeGroot–Marshak pricing rule, and one direct question in which we explicitly ask for predictions, under a quadratic incentive scheme. Both rules have the theoretical property that if a subject is risk neutral he or she should optimally reveal the “true” expected value of his or her subjective probability distribution over Group B behavior, regardless of the form of this distribution.³ From a behavioral perspective, however, there may be arguments in favor of using either of the mechanisms: while the quadratic scoring rule has the disadvantage that the subjects' incentives are essentially flat in the neighborhood of the optimal response, the bidding mechanism (which gives monetary incentives that are linear in the deviation from the optimal response) is considerably more complicated.

The experimental results under both mechanisms reveal a systematic and significant prediction error: although the subjects almost always correctly predict the action that is chosen with higher frequency, their elicited beliefs are distorted towards the equiprobable belief (or uniform prior) which assigns equal probabilities to both choices in the action set of Group B. On average, subjects are conservative in their probability assessments.^{4,5} Also, we observe that the quadratic incentive rule consistently yields more accurate predictions than the bidding mechanism.

The following section presents our experimental design and procedures. Section 3 contains the main results, and Section 4 offers a brief summary.

² Group A subjects knew that Group B subjects had already been paid off, as they were truthfully told that Group B had participated in an experiment in the past.

³ For a discussion of both mechanisms see Kadane and Winkler (1988), who also address the question of response distortions in the presence of more general risk attitudes. Rabin (2000), however, illustrates the difficulties that arise when risk aversion is allowed in analyses of experiments with low monetary incentives.

⁴ Following Edwards (1968) we use conservatism as a term for belief distortions towards the uniform prior.

⁵ In the context of normal-form games, Stahl and Wilson (1994, 1995), and Costa-Gomes et al. (2000) find that a substantial proportion (20–30%) of the subjects consistently play the best response against the equiprobable belief. Our experimental results confirm this tendency of conservatism in beliefs about others' behavior, using more explicit elicitation methods. An experimental procedure similar to ours has been used by Hsee and Weber (1997) who asked subjects to predict risk preferences of others and find that subjects systematically predict others to be less risk averse than they are themselves.

2. Experimental design and procedures

The experiment was conducted in summer 1997 and fall 1999 at Humboldt University in Berlin, in four sessions with altogether 71 Group A subjects. The participants were students from various fields, most of them undergraduates in economics and business administration. Subjects were sitting in large lecture rooms with enough space between them to rule out any attempts of communication. All subjects received a fee of DM 20 (about US\$ 13 at that time) on top of their performance-dependent earnings.

Each session consisted of four different tasks, the order of which was varied between two treatment groups. In the following we describe the order of tasks for Group A1.⁶

2.1. Part 1

Subjects were made familiar with the Becker–De Groot–Marshak (BDM) mechanism which is an incentive compatible mechanism to buy (or sell) goods.⁷ The instructions carefully explained why it is optimal to bid one's true valuation under this mechanism. Then we sold bags of "gummi bears" to the subjects using the BDM mechanism. After submitting their bids subjects could observe all steps of price determination and allocation of the gummi bears.

2.2. Part 2

Subjects received a set of instructions we had used in the previously conducted lottery-choice experiment of Group B (see Appendix B).⁸ They were truthfully told that these were the original instructions from a separate experiment and they were given some basic information about the subjects who had participated in it: that they were 69 pupils in grade 12 of a German 'Gymnasium', 43 of whom were female, that their average age was 18.1 years, and that they had an average monthly budget of DM 311. Also, the Group A subjects received graphical descriptions of 8 of the 16 binary-choice tasks that the subjects of Group B had been confronted with. Each of these eight tasks was then used to create a new prospect for the present experiment, as follows: Group A subjects were told that we would pick one of the Group B subjects at random. Depending on this subject's decision in a given task (for left or right) the according prospect would either pay off DM 5 or 2. Then we sold these new prospects to the present subjects using the BDM mechanism. An example is illustrated in Fig. 1. Here a Group A subject would receive DM 5 if the randomly chosen Group B subject had chosen the left lottery. He or she would earn DM 2 if the Group B subject had chosen the right lottery. With this information he or she had to decide how much he or she would be willing to pay for the prospect. An overview of the eight lottery-choice tasks involved

⁶ The complete instructions for Group A1 can be found in Appendix A.

⁷ The mechanism we used, which was first proposed by Becker et al. (1964), works as follows: potential buyers decide upon their bid prices. Then the actual price is randomly determined by drawing a price from a given distribution (the support of which must be large enough). The good is sold by the experimenter at the randomly determined price to all buyers who have submitted a higher bid price. In this setup submitting one's true willingness to pay is a dominant strategy.

⁸ This experiment is documented in more detail and analyzed in Huck and Weizsäcker (1999b).

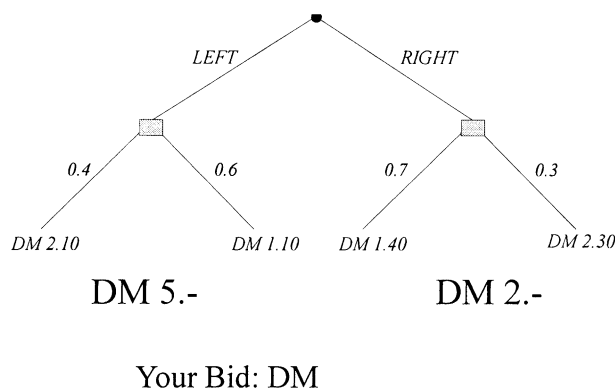


Fig. 1. Example of a lottery sold to Group A subjects (task 1).

is given in Table 1, where $(x_{t1}, q_{t1}; x_{t2}, q_{t2})$ is to be read as a prospect paying off DM x_{tj} with probability q_{tj} , $t = 1, \dots, 8$, $j = 1, 2$.

Given the assumptions of rational bidding and risk neutrality a bid reveals a subject's expectation about choice frequencies in Group B. More precisely, subject i 's subjective expected value of the relative frequency of Group B subjects choosing the alternative leading to a payoff of DM 5 in task t is $(R_{it}^{\text{BDM}} - \text{DM } 2) / (\text{DM } 5 - \text{DM } 2)$, if R_{it}^{BDM} is i 's bid in t .

2.3. Part 3

In this part we elicited subjects' expectations in an alternative, more straightforward manner. Subjects in Group A were again confronted with the same eight tasks of Group B, but this time we asked them directly how they thought the other group had decided, i.e. they had to write down how many of the 69 subjects of Group B they thought had decided for left. To give the Group A subjects proper incentives for submitting their true expected values we used the quadratic incentive rule

$$\pi_i = A - k \sum_{t=1}^8 (R_{it}^{\text{dir}} - P_t)^2, \quad (1)$$

Table 1
Pairs of lotteries

Task t	Left lottery	Right lottery
1	(2.10, 0.4; 1.10, 0.6)	(2.30, 0.3; 1.40, 0.7) ^a
2	(1.96, 0.24; 1.68, 0.34; 1.18, 0.42)	(1.93, 0.23; 2.26, 0.28; 1.29, 0.49) ^a
3	(2.29, 0.41; 1.39, 0.59) ^a	(1.51, 0.21; 2.21, 0.36; 1.11, 0.43)
4	(0.53, 0.36; 1.31, 0.64)	(0.98, 0.44; 1.68, 0.56) ^a
5	(1.10, 0.4; 1.70, 0.6)	(0.80, 0.2; 1.80, 0.8) ^a
6	(1.59, 0.27; 0.99, 0.31; 1.91, 0.42) ^a	(0.67, 0.26; 1.64, 0.74)
7	(1.92, 0.3; 0.73, 0.7) ^a	(0.98, 1)
8	(1.20, 0.26; 1.30, 0.27; 0.80, 0.47) ^a	(1.10, 0.31; 1.20, 0.32; 0.60, 0.37)

^a Lotteries with greater expected values.

Table 2
The two treatments

	Group A1	Group A2
No. of subjects	44	27
Part 1	BDM practice	Direct mechanism
Part 2	BDM mechanism	BDM practice
Part 3	Direct mechanism	BDM mechanism
Part 4	Own decisions	Own decisions

where π_i is i 's payoff from this part, A and k are constant, P_t is the absolute number of Group B subjects choosing left in task t , R_{it}^{dir} is i 's response under this mechanism, and $R_{it}^{\text{dir}}, P_t \in \{0, \dots, 69\}$. We chose $A = \text{DM } 10$ and $k = \text{DM } 0.002$. Under this quadratic incentive rule a rational and risk neutral individual will set R_{it}^{dir} to equal his or her subjective expected value of the unknown P_t , regardless of his or her subjective probability distribution over the possible realizations of P_t .⁹

2.4. Part 4

In the last part of the experiment subjects were confronted with the same eight tasks again. However, this time they did not have to think about Group B, but rather had to decide between the lotteries themselves. They received payoffs for four randomly selected tasks. After all decisions had been made we invited all subjects to stay for some more minutes to watch how all the chance moves were carried out.¹⁰ Then payments were calculated and subjects could pick up their money the following day.

In order to control for learning effects we reversed the order of tasks for Group A2 subjects. Table 2 summarizes the design.

3. Experimental results

Table 3 presents the aggregate data of the experiment. The table contains the average values of all actual and estimated relative choice frequencies. It shows for each task (a) the actual proportion of subjects in Group B who chose left, (b) the average belief of Group A subjects thereof as elicited by the BDM mechanism, (c) the same thing as elicited by the direct mechanism, and finally (d) the proportion of subjects in Group A who chose left. For (b) and (c) we also add standard deviations (in parentheses). A single (double) asterisk after the standard deviation indicates that the mean estimate is significantly different at the 5% (1%) level from the true Group B choice frequency and that the deviation is in line with conservatism. Minus signs indicate significant deviations that are not in line with conservatism. Fig. 2 plots the average belief data.

⁹ This property was not explained to the subjects.

¹⁰ This had been announced before the experiment and we had made sure that at least one subject would stay to monitor the procedures.

Table 3

Aggregate results: average actual and predicted frequencies of choosing left^a

Task t	1	2	3	4	5	6	7	8
(a) Choice frequencies in Group B	0.159	0.101	0.826	0.029	0.420	0.638	0.522	0.725
(b) Average beliefs under BDM								
In Group A1	0.312 (0.301)**	0.358 (0.258)**	0.590 (0.244)**	0.241 (0.207)**	0.409 (0.299)	0.525 (0.287)*	0.515 (0.309)	0.689 (0.252)
In Group A2	0.624 (0.222)**	0.075 (0.116)	0.870 (0.168)	0.519 (0.221)**	0.189 (0.207)– –	0.829 (0.162)–	0.297 (0.233)– –	0.257 (0.200)– –
In both groups	0.430 (0.312)**	0.250 (0.255)**	0.697 (0.256)**	0.347 (0.251)**	0.326 (0.287)– –	0.641 (0.287)	0.432 (0.300)–	0.524 (0.314)**
(c) Average beliefs under direct mechanism								
In Group A1	0.246 (0.194)**	0.313 (0.171)**	0.628 (0.184)**	0.206 (0.145)**	0.397 (0.248)	0.583 (0.227)	0.446 (0.275)	0.595 (0.207)**
In Group A2	0.189 (0.136)	0.253 (0.171)**	0.657 (0.221)**	0.131 (0.098)**	0.301 (0.207)– –	0.691 (0.164)	0.515 (0.275)	0.657 (0.168)*
In both groups	0.225 (0.176)**	0.291 (0.173)**	0.639 (0.198)**	0.178 (0.134)**	0.360 (0.236)–	0.624 (0.211)	0.472 (0.275)	0.618 (0.194)**
(d) Choice frequencies								
In Group A1	0.045	0.045	0.863	0.045	0.318	0.772	0.568	0.841
In Group A2	0.037	0.000	0.852	0.000	0.148	0.889	0.778	0.889

^a Standard deviations in parentheses. A double (single) asterisk indicates that the predicted frequency is significantly different from the Group B choice frequency on the 1%-level (5%-level) and that this difference is in line with conservatism. Minus signs indicate significant differences in the same manner for cases in which the differences are not in line with conservatism (one-sample two-sided t -tests).

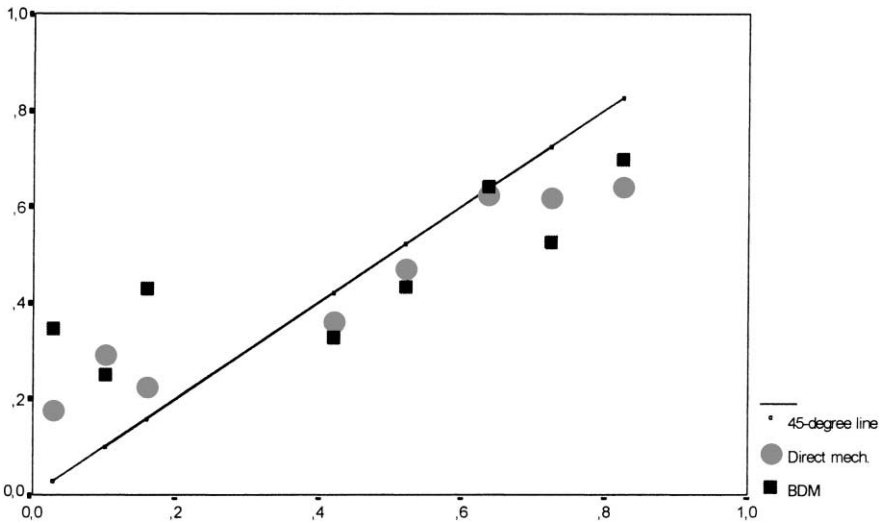


Fig. 2. Average predicted frequencies of choosing left. Group B's relative choice frequencies in the eight tasks are depicted on the horizontal axis. On the vertical axis, for each task the estimates under the BDM mechanism (squares) and the estimates under the direct mechanism (disks) are given.

Several observations are in order.

1. While there are essentially no differences between Groups A1 and A2 under the direct mechanism, the differences under the BDM mechanism are large and highly significant.¹¹
2. Estimates under the direct mechanism are more accurate.¹²
3. While subjects, on average, correctly predict which of the two decisions is more frequently chosen, their beliefs are systematically distorted towards one half.
4. The belief distortions cannot be explained by Group A's own behavior, i.e. we do not observe any indication of a false-consensus effect which occurs when subjects overestimate the probability that others act like themselves (see, e.g. Mullen et al. (1985)).¹³

¹¹ While it is impossible to reject, for any of the eight tasks, the null hypothesis that the direct mechanism responses of both groups come from the same distributions, each of the analogous eight null hypotheses for the BDM mechanism responses can be rejected at the 1% level (two-tailed Mann–Whitney U tests).

¹² To establish this claim we use a simple quadratic scoring rule, measuring the distance between average beliefs and Group B choice frequencies. While the BDM scores -0.27 the direct mechanism does much better with -0.12 .

¹³ Rather, Group A (university students) subjects reveal a more extreme decision pattern than Group B subjects (high school students): their relative choice frequencies typically are closer to either 0 or 1. This mirrors the observation that the university students in Group A chose more in accordance with expected value maximization than Group B subjects.

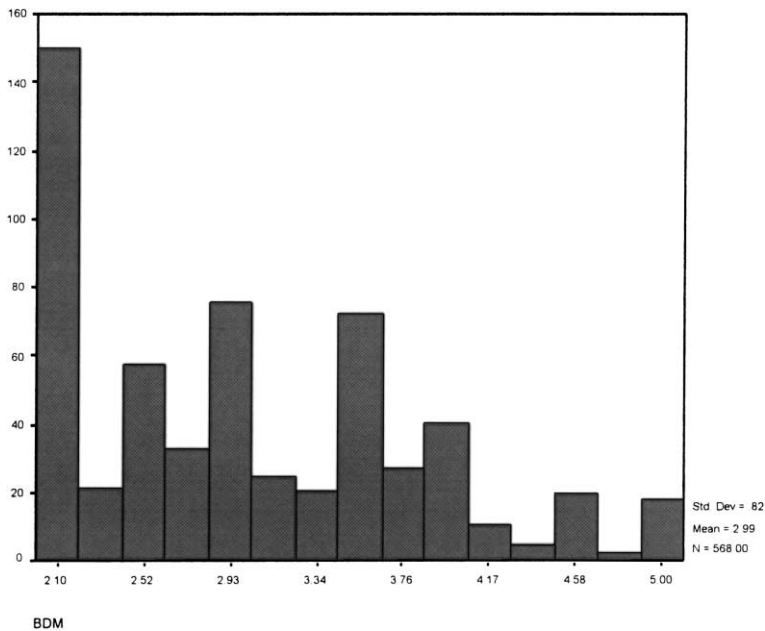


Fig. 3. Responses under BDM.

Observations 1 and 2 suggest that the BDM mechanism is less reliable than the direct mechanism: it is less robust and the elicited beliefs are farther off the true frequencies. This is underlined by two additional observations. First, the BDM mechanism yielded systematic distortions in that 17.8% of the bids equal the lowest possible response of DM 2, suggesting that large parts of the responses cannot reflect the subjects' true expectations about Group B behavior (3.0% of the responses equal DM 5). Of the responses under the direct mechanism, only 3.0% lie on either of the bounds of the response range. (See also Figs. 3 and 4 which show the distributions of responses under the two mechanisms, aggregated over all tasks. As these figures illustrate, the BDM data also tend to cluster in the interior of the response range.) Second, subjects frequently give the same response under the BDM mechanism for several tasks while they give differing responses for the same set of tasks under the direct mechanism: under the BDM mechanism, there are 275 instances in which a subject gives an identical response for a pair of tasks for which he or she gives differing responses under the direct mechanism. The reverse imbalance occurs in only 68 cases, i.e. the responses under the direct mechanism are much more differentiated.

For the more reliable responses under the direct mechanism we find that the belief distortions towards the uniform prior are consistent and significant. There is just one task where average beliefs are significantly distorted in the other direction, namely task 5 where the true frequency (0.42) is close to 0.5. For the only task with the true frequency even closer to 0.5 (task 7, where it is 0.522) we do not find a significant difference in either direction, whereas in five of the six remaining tasks the distortion towards 0.5 is significant on the

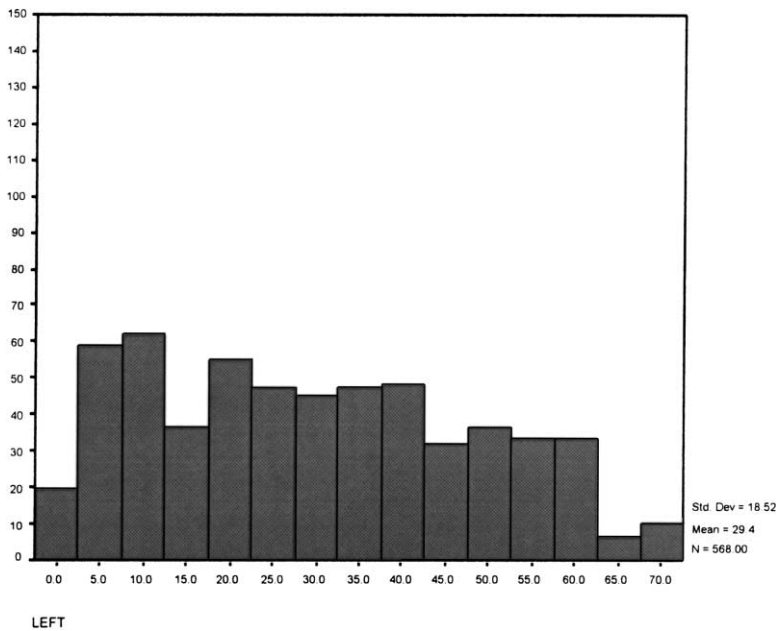


Fig. 4. Responses under direct mechanism.

1% level. In all, we find considerable evidence for conservatism in subjects' beliefs about what others do.¹⁴

4. Summary

The paper presents an experiment in which subjects are asked to give estimates about the behavior of other subjects who had participated in a separate lottery-choice experiment. Two different elicitation mechanisms were used. While the direct mechanism with a quadratic scoring rule yields better predictions than the competing bidding mechanism does, there is a significant distortion of the estimates towards the uniform prior even if the “superior” direct mechanism is used.

Acknowledgements

We are indebted to Jörg Breitung, Dorothea Kübler, Jörg Oechssler, and David Grether who acted as editor, for many helpful comments and suggestions. Financial support from the Deutsche Forschungsgemeinschaft (DFG) is gratefully acknowledged.

¹⁴ In Huck and Weizsäcker (1999a) we also estimate a linear model of the predicted choice frequencies. Assuming that the mean choice predictions are linear in the actual choice frequencies, we estimate the relative weights of the actual choice probabilities and of the uniform prior in the determination of the stated expectations of Group A. The regression analysis confirms that the weight put on the uniform prior is substantial and highly significant.

Appendix A. Group A instructions

Welcome to our experiment in which you will make decisions in a number of choice tasks during the next 90 min. You will be paid by us for your participation. First you receive a fixed payment of DM 20. Depending on your decisions you will receive further payments in the course of the experiment, so you can, in addition to the DM 20, earn further money later on.

Before entering the first round of the experiment, a couple of basic rules: it is important that you do not talk to each other from now on. You are supposed to only consider your own set of material. In case you have questions, please raise your hand. We will then come to your seat and answer them privately. Whoever does not comply with these rules will, unfortunately, be excluded from all payoffs.

Now let us turn to the first phase of the experiment, in which you get to know the so-called BDM mechanism (named after economists G. Becker, M. DeGroot, and J. Marshak). Using this mechanism, you can buy several goods. For illustration of the mechanism we offer each of you to buy a bag of “Haribo Goldbären” from us. If the purchase takes place, we will hand out the bag to you after the experiment. The procedure is as follows: you silently decide on a price (minimum DM 0.50, maximum DM 1.50) at which you are willing to buy the bag of gummi bears, and write down the price on a prepared form. After we collected all forms, we will draw a chip from an urn containing 100 chips. Each chip shows a number between 0.51 and 1.50 (that is, one chip shows 0.51, one 0.52, and so on, until 1.49, and finally 1.50). The chip that was drawn determines the price at which we sell the gummi bears. For example, if we draw a chip showing 0.79, we will sell the bags at exactly DM 0.79, to those of you who are willing to pay at least DM 0.79. Hence, even if you wrote down a price of, say, DM 1.23, you will buy a bag from us at DM 0.79. If you wrote down a price below DM 0.79, you do not purchase the gummi bears.

This procedure may, admittedly, sound somewhat complicated. However, using this mechanism has a considerable advantage: it should make it easy for you to decide on how much money you bid for one bag of gummi bears, due to the following reason: suppose you find that a bag is worth exactly DM 1.14 to you. If you saw it in a store at a lower price, you would happily be willing to buy it, at a price of DM 1.14 you would still be willing to purchase, and prices of DM 1.15 and above are too expensive. What should you do in this experiment? It is rather simple to see that the best thing you can do is to bid exactly DM 1.14!

Why? Imagine you bid less instead, say, exactly DM 1. What could happen? For example, the randomly drawn price could lie below DM 1. In this case, nothing would change for you; at both bids (1.14 and 1.00) you would get the bag at a low price and would supposedly be satisfied with this. Also, the drawn price could lie above DM 1.14. Again, your decision to bid DM 1 would not change anything: you would be glad not to be forced to buy the bag in any case. However, imagine that we draw a price between DM 1.00 and DM 1.14, for example DM 1.05. In this case you would certainly not be glad, because due to the fact that you bid below DM 1.05, you will not purchase the bag; although you would like to buy, since it is worth DM 1.14 to you. Therefore, you see that it cannot pay off to you to bid a price lower than what the bag is actually worth to you. In a similar manner, one can see that it is also useless to bid a price above your “true valuation”: suppose you bid DM 1.30. Should the price drawn lie below DM 1.14 or above DM 1.30, you would neither gain

nor loose. Should, however, the price lie in between, you would be forced to buy the bag at price that you would not be willing to pay.

No further questions? Then let us proceed to selling the gummi bears. For each of you we prepared a bag. Now each of you may want to think about how much such a bag is actually worth to him or her. In any case please decide on a bid for one bag now. You will be given a form to fill in the bid in a second. Consequently, we will determine the price by drawing a chip, and write down who of you purchases at this price. Those of you who do buy will get a bag, at the according price, for your way home.

Apart from this sheet of instructions you also received a card with your code number. Please fill in this code number on each of the forms that you receive during the experiment. Also please keep the card at a secure place, as you will receive your monetary payoffs only if you return the card to us. This procedure assures your anonymity. We will only be able to relate all your decisions to a code number, and not to you in person.

[Decision Sheet I, gummi bear bids.]

In the following we ask you to consider a separate experiment that we conducted at a ‘Gymnasium’ in Neu-Isenburg, close to Frankfurt/Main, on 31 May, last year. You receive the original instructions used in this experiment, original decision sheets that the students had to fill in, and (on this sheet, below) some information on the participating subjects. The reason for our asking to read this material is that you will have to make decisions that are related to this experiment conducted last year. We will be more precise on this later on. For now, here is some information on the participating subjects of the experiment:

- The participants of the experiment were the students of all classes in grade 12 of the ‘Goethe-Gymnasium’ in Neu-Isenburg, close to Frankfurt/Main. In total 69 subjects participated, 43 of whom were female.
- The average age of the participants was 18.1 years.
- On average, subjects had at their disposition a budget of DM 311 (excluding expenses for clothing).

[Instructions of Group B, see Appendix B, and a separate sheet depicting the eight decision tasks.]

Now we proceed to the next part of the experiment. This time you will be asked to make a number of decisions.

By reading the material distributed a couple of minutes ago, you could and can become familiar with an experiment that we conducted at the ‘Goethe-Gymnasium’ in Neu-Isenburg last year. Now this experiment will be of some importance for your own decisions.

Consider the lotteries displayed on Decision Sheet II. Each of these lotteries has two possible outcomes: DM 2 and DM 5. How do they work? Put differently, in which case does someone playing this lottery receive DM 2, and in which case DM 5? The answer is simple: we determine the payoff of the lottery by randomly drawing one of the 69 subjects from Neu-Isenburg. If this subject chose *left* in his or her decision task, you receive the amount of money given under *left*. If he or she chose *right*, you receive the amount given under *right*. Hence, for the first of the lotteries, in the upper left corner, the following payoff rule applies: one of the 69 subjects is randomly chosen; if he or she chose his or her right alternative, *right*, you receive DM 2. If he or she chose *left*, you receive DM 5. The actual realizations of the random draws that determined the payoffs of the subjects in the

Neu-Isenburg experiment are irrelevant for you. The same rules apply to the other seven lotteries on your decision sheet.

In the following you can buy these lotteries from us, by using the BDM mechanism described above. If you buy a lottery, you pay the randomly determined price (which, since you buy, cannot be greater than your bid). After that, as explained before, we draw one of the 69 subjects. His or her decision will determine whether you receive DM 2 or DM 5. If the purchase does not take place, you pay nothing and do not receive a payoff from this lottery.

We ask you to indicate, for each of the eight lotteries, the price that you are willing to pay for the lotteries, on Decision Sheet II.

The game played in this round has, therefore, the following rules, described for the example of the first lottery:

- You write down the price you are willing to pay, i.e. *your bid*, below the lottery. Your bid has to be at least DM 2 and at most DM 5. You do not have to round numbers, i.e. all possible bids on a grid with the coarseness of one Pfennig are allowed.
- After collecting all bids we draw a random price from an urn containing the 300 possible prices from DM 2.01 to DM 5.00.
- If your bid is greater than or equal to the price drawn, you buy the lottery; i.e. you pay the randomly drawn price. Otherwise the lottery remains ours and is irrelevant for you.
- If you buy the lottery we make a draw from a second urn, with integers 1–69, to determine one of the subjects of last year's experiment.
- If this subject chose *left*, you receive DM 5 (for the example of the first lottery). If he or she chose *right*, you receive DM 2.

All payments (from us to you or vice versa) will be added and made in the following days. Some of you will be present to supervise the drawings from the urns.

Please also consider the following:

- The subjects of last year's experiment are, of course, paid off already. Thus, your decision today is irrelevant for them. Also, the subjects in Neu-Isenburg did not know anything about our experiment today. On their decision sheets, your possible payoffs were not given (DM 2/5).
- Recall that the BDM mechanism is constructed such that you have no incentive to give a different bid from what a lottery is *actually worth* to you.
- Please do not forget to enter your code number on the decision sheet.

[Decision Sheet II, eight lotteries as given in Fig. 1.]

Now let us turn to the next part of the experiment.

In the last round you had to bid for the lotteries which were influenced by the decisions of last year's subjects. Now we ask you to give explicit estimates on how many of the 69 subjects chose the left or the right alternatives in the lotteries.

On the distributed Decision Sheet III you again see the decision tasks of the students who participated in the experiment in Neu-Isenburg. Below each of the decision trees please indicate your estimate of the (absolute) number of subjects that chose *left* in the experiment. If, for example, you assume that exactly one third of the participants chose *left* in a given decision task, please give, below the according tree, the number 23 as your response. Please only use integers from the interval [0; 69].

As before, monetary payoffs are made in this part of our experiment, as follows: as an initial endowment you receive a one-time payment of DM 10. (This is independent of your show-up fee of DM 20, and it is not to be confused with the latter.) However, deductions will be made from these DM 10, depending on the accuracy of your estimates. From last year's experiment we of course know the exact numbers of subjects choosing *left* and *right* in each task.

The difference between your estimate and the actual number will be squared (multiplied by itself) and divided by 100, for each of the eight tasks. The resulting values will be multiplied by a factor of 0.2 and deducted from your initial endowment. Assume for example that your response, for a given task, is the number 23, whereas in fact only 20 of the 69 students chose *left*. In this case you loose, due to the difference of three subjects, the amount of DM $0.2 \times (23 - 20)^2 / 100 = \text{DM } 0.018$, in words one point eight Pfennigs. Greater differences from the actual numbers can, of course, cause greater losses; if you, for example, miss the true number by 10 subjects, 20 Pfennigs will be deducted, a misprediction of 20 subjects makes you loose 80 Pfennigs, and a misprediction of 30 subjects makes you loose DM 1.80.

The amount remaining after all deductions will be paid off to you together with your gains from the other parts of the experiment. If you cannot earn money in this round, due to gross mispredictions of the Neu-Isenburg subjects, your losses will be deducted from the gains of the other rounds.

On Decision Sheet III, now please enter your estimates below the decision trees. Please give exactly one response for each task.

[Decision Sheet III, lines for responses below the eight decision tasks.]

We now turn to a further part of the experiment. You again receive a decision sheet, the contents of which you are quite familiar with by now. Depicted are the same eight pairs of lotteries that the Neu-Isenburg students had to choose from and that you already know rather well. We now want you to make a decision between the lotteries themselves — exactly in the same manner as the students of last year's experiment were supposed to do. Again you can earn money. We will randomly draw four pairs of lotteries that will actually be paid off. That is, after we selected the four pairs of lotteries, we will, in accordance to your decisions, make the random draw of the left or the right lottery in each of the tasks. The resulting monetary payoffs will be added to your total payoff.

[Decision Sheet IV, eight decision tasks.]

Appendix B. Group B instructions

Welcome to our experiment which will demand approximately 90 minutes of your time. During the experiment, we will ask you to make a number of decisions. The kind of decisions you will have to make is explained below. First, it is important to know that you will have the opportunity to earn money. Thus, if we mention *payoffs* in the following, we will always be talking about real German Marks (DM). Together with these instructions you received a code card. **Please do not forget to indicate your code number on all forms you receive from us.** Please do also remember not to throw away your code card after the experiment. Only if showing your code card you will be given your payoff. In addition, the code cards guarantee your full anonymity during and after the experiment.

It is furthermore important that the data we obtain by conducting the experiment are independent. This means: please do not talk to your neighbors, look at your decisions forms only, do what *you* consider to be reasonable and nothing else. If you do not stick to these rules we will have to exclude you from the experiment. On your desk, you will only have a pen and the material we are going to hand out.

Now we turn to the instructions:

During the experiment you will have to make 16 decisions. Each decision consists of the choice between two *lotteries*. A lottery can have (as you probably know from public lotteries) several outcomes, each outcome corresponding to a certain probability. Depending on which of the outcomes is realized, you will receive a payoff. A simple lottery could, for example, consist of the following: you receive a payoff of DM 1.20 with a probability of 0.6 (i.e. 60%, or in 6 of 10 possible cases, respectively) and a payoff of DM 2.00 with the remaining probability of 0.4 (i.e. 40%, in 4 of 10 possible cases). This lottery can be depicted like this:

[Small tree with two branches, depicting the lottery described above.]

In this 'tree', the left branch represents the event that occurs with probability 0.6, inducing a payoff of DM 1.20 for you, while the right branch represents the event occurring with probability 0.4 leading to a payoff of DM 2.00. Of course, it is also possible to represent lotteries with more than two possible events analogously.

Your task will be to choose between two lotteries, which we will call *left* and *right*. You will have to make 16 of these decisions. Each of them will be presented like this:

[Figure with a tree like the one in Section 2. Left lottery (1.20, 0.6; 2.00, 0.4), right lottery (200.00, 0.7; 120.00, 0.3).]

You then have to choose one of the lotteries. This lottery will be decisive for your payoff. Suppose you had chosen the left lottery of the example above (which would not have been very clever). Then we would take an urn with 100 chips, 60 of which represent a payoff of DM 1.20, and 40 represent a payoff of DM 2.00. The actually chosen chip would determine your payoff.

Since you have to make 16 decisions we will consequently draw chips from 16 urns, and, of course, you will be paid according to the sum of all 16 outcomes. (Actually, some of the participants will be witnesses when we draw the chips, so you can be sure that we will not manipulate the results).

On your decision forms you will indicate your decisions by making a circle around *left* or *right*, i.e. per pair of lottery you have to make exactly one circle.

Please do not forget to keep quiet during the experiment and to make your decisions carefully and without any haste.

If you have a question raise your hand. We will come to your table.

Have fun!

References

- Becker, G.M., De Groot, M.H., Marshak, J., 1964. Measuring utility by a single-response sequential method. *Behavioral Science* 9, 226–232.
- Costa-Gomes, M., Crawford, V., Broseta, B., 2000. Cognition and behavior in normal-form games: an experimental study. *Econometrica*, in press.

- Croson, R.T.A., 2000. Thinking Like a Game Theorist: Factors Affecting the Frequency of Equilibrium Play. Mimeo, University of Pennsylvania.
- Dufwenberg, M., Gneezy, U., 2000. Measuring beliefs in an experimental lost wallet game. *Games and Economic Behavior* 30, 163–182.
- Edwards, W., 1968. Conservatism in human information processing. In: Kleinmuntz, B. (Ed.), *Formal Representation of Human Judgement*. New York, pp. 17–52.
- Hsee, C.K., Weber, E.U., 1997. A fundamental prediction error: self-other discrepancies in risk preference. *Journal of Experimental Psychology* 126, 45–53.
- Huck, S., Weizsäcker, G., 1999a. Do players correctly estimate what others do? The Role of Underconfidence in Beliefs. Mimeo, Humboldt University, Berlin.
- Huck, S., Weizäcker, G., 1999b. Risk, complexity, and deviations from expected value maximization. *Journal of Economic Psychology* 20, 699–715.
- Kadane, J.B., Winkler, R.L., 1988. Separating probability elicitation from utilities. *Journal of the American Statistical Association* 83, 357–363.
- Mullen, B., Atkins, J.L., Champion, D.S., Edwards, C., Hardy, D., Story, J.E., Vanderklok, M.T.H.E., 1985. False consensus effect: a meta-analysis of 115 hypothesis tests. *Journal of Experimental Social Psychology* 21, 262–283.
- Nyarko, Y., A. Schotter, A., 1999. An experimental study of belief learning using real beliefs. New York University, Mimeo.
- Offerman, T., Sonnemans, J., Schram, A., 1996. Value orientations, expectations and voluntary contributions in public goods. *Economic Journal* 106, 817–845.
- Rabin, M., 2000. Risk aversion, diminishing marginal utility and expected-utility theory: a calibration theorem. *Econometrica* 68, 1281–1292.
- Stahl, D.O., Wilson, P.W., 1994. Experimental evidence on players' models of other players. *Journal of Economic Behavior and Organization* 25, 309–327.
- Stahl, D.O., Wilson, P.W., 1995. On players' models of other players: theory and experimental evidence. *Games and Economic Behavior* 10, 218–254.